

Event Codes.

Event Codes

Event 101,102,103,104,105

Event message:

- **Grid incident**

Explanation:

The grid voltage or grid impedance at the connection point of the inverter is too high. The inverter has disconnected from the utility grid.

Corrective measures:

- Check whether the grid voltage at the connection point of the inverter is permanently in the permissible range.

If the grid voltage is outside the permissible range due to local grid conditions, contact the grid operator. The grid operator must agree with an adjustment of the voltage at the feed-in point or with a change of the monitored operating limits.

If the grid voltage is permanently within the permissible range and this message is still displayed, contact the Service.

Event 202,203,205,206

Event message:

- **Grid incident**

Explanation:

The utility grid has been disconnected, the AC cable is damaged or the grid voltage at the connection point of the inverter is too low. The inverter has disconnected from the utility grid.

Corrective measures:

- Ensure that the miniature circuit breaker is switched on.
- Ensure that the AC cable is not damaged and that it is connected correctly.

- Ensure that the country data set has been configured correctly.
- Check whether the grid voltage at the connection point of the inverter is permanently in the permissible range.

If the grid voltage is outside the permissible range due to local grid conditions, contact the grid operator. The grid operator must agree with an adjustment of the voltage at the feed-in point or with a change of the monitored operating limits.

If the grid voltage is permanently within the permissible range and this message is still displayed, contact the Service.

Event 301

Event message:

- **Grid incident**

Explanation:

The ten-minute average value of the grid voltage is no longer within the permissible range. The grid voltage or grid impedance at the connection point is too high. The inverter disconnects from the utility grid to maintain power quality.

Corrective measures:

- During the feed-in operation, check whether the grid voltage at the connection point of the inverter is permanently in the permissible range.

If the grid voltage is outside the permissible range due to local grid conditions, contact the grid operator. The grid operator must agree with an adjustment of the voltage at the feed-in point or with a change of the monitored operating limits.

If the grid voltage is permanently within the permissible range and this message is still displayed, contact the Service.

Event 302

Event message:

- **Active power limited AC voltage**

Explanation:

The inverter has reduced its power due to a too-high grid voltage to ensure grid stability.

Corrective measures:

- If possible, check the grid voltage and observe how often fluctuations occur. If fluctuations occur frequently and this message is displayed often, contact the grid operator and request approval to change the operating parameters of the inverter. If the grid operator gives approval, discuss any changes to the operating parameters with the Service.

Event 401

Event message:

- **Grid incident**

Explanation:

The inverter has disconnected from the utility grid. A stand-alone grid or a very large change in the grid frequency was detected.

Corrective measures:

- Make sure there is no power outage or work being done on the utility grid and contact a grid operator if necessary.
- Check the grid connection for significant short-term frequency fluctuations.

Event 404

Event message:

- **Grid incident**

Explanation:

The inverter has disconnected from the utility grid. A stand-alone grid or a very large change in the grid frequency was detected.

Corrective measures:

- Make sure there is no power outage or work being done on the utility grid and contact a grid operator if necessary.

- Check the grid connection for significant short-term frequency fluctuations.

Event 501, 502, 503

Event message:

- **Grid incident**

Explanation:

The grid frequency is not within the permissible range. The inverter has disconnected from the utility grid.

Corrective measures:

- If possible, check the grid frequency and observe how often fluctuations occur.

If fluctuations occur frequently and this message is displayed often, contact the grid operator and request approval to change the operating parameters of the inverter.

If the grid operator gives approval, discuss any changes to the operating parameters with the Service.

Event 507

Event message:

- **Active power limit AC frequency**

Explanation:

The inverter has reduced its power due to a too-high grid frequency to ensure grid stability.

Corrective measures:

- If possible, check the grid frequency and observe how often fluctuations occur. If fluctuations occur frequently and this message is displayed often, contact the grid operator and request approval to change the operating parameters of the inverter. If the grid operator gives approval, discuss any changes to the operating parameters with the Service.

Event 601

Event message:

- **Grid incident**

Explanation:

The inverter has detected an excessively high proportion of direct current in the grid current.

Corrective measures:

- Check the grid connection for direct current.
- If this message is displayed frequently, contact the grid operator and check whether the monitoring threshold on the inverter can be raised.

Event 701

Event message:

- **Frequency not permitted**
- **Check parameter**

Explanation:

The grid frequency is not within the permissible range. The inverter has disconnected from the utility grid.

Corrective measures:

- Check the AC wiring from the inverter to the feed-in meter.
- If possible, check the grid frequency and observe how often fluctuations occur.

If fluctuations occur frequently and this message is displayed often, contact the grid operator and request approval to change the operating parameters of the inverter.

If the grid operator gives approval, discuss any changes to the operating parameters with the Service.

Event 901

Event message:

- **PE connection missing**
- **Check connection**

Explanation:

The grounding conductor is not correctly connected.

Corrective measures:

- Ensure that the grounding conductor is correctly connected.

Event 1001

Event message:

- **L / N swapped**

Explanation:

Incorrect installation of the L / N connection.

Corrective measures:

- Check the AC wiring from the inverter to the feed-in meter.
- Check the AC voltages on the inverter connection.
- If this message is still displayed, contact the Service.

Event 1101

Event message:

- **Installation error**

- **Check connection**

Explanation:

A second line conductor is connected to N.

Corrective measures:

- Connect the neutral conductor to N.

Event 1302

Event message:

- **Waiting for grid voltage**
- **Grid connection installation failure**
- **Check grid and fuses**

Explanation:

L or N not connected.

Corrective measures:

- Ensure that the line conductors are connected.
- Ensure that the miniature circuit breaker is switched on.
- Ensure that the AC cable is not damaged and that it is connected correctly.

Event 1501

Event message:

- **Reconnection fault grid**

Explanation:

The changed country standard or the value of a parameter you have set does not correspond to the local requirements. The inverter cannot connect to the utility grid.

Corrective measures:

- Ensure that the country data set has been configured correctly. To do this, select the parameter **Set country standard** and check the value.

Event 3302, 3303

Event message:

- **Unstable operation**

Explanation:

There is not enough power at the DC input of the inverter for stable operation. The inverter cannot connect to the utility grid.

Corrective measures:

- Ensure that the PV module is dimensioned correctly.
- Ensure that the PV module is not covered by snow or otherwise shaded.
- Ensure that the PV module is free of defects.

Event 3501

Event message:

- **Insulation failure**
- **Check generator**

Explanation:

The inverter has detected a ground fault in the PV module.

Corrective measures:

- Check the PV system for ground faults.

Event 3601

Event message:

- **High discharge current**
- **Check generator**

Explanation:

The leakage current of the inverter and the PV module is too high. There is a ground fault, a residual current or a malfunction.

The inverter interrupts feed-in operation immediately after exceeding a threshold. When the fault is eliminated, the inverter automatically reconnects to the utility grid.

Corrective measures:

- Check the PV system for ground faults.

Event 3701

Event message:

- **Residual current too high**
- **Check generator**

Explanation:

The inverter has detected a residual current through brief grounding of the PV module.

Corrective measures:

- Check the PV system for ground faults.

Event 3801, 3802, 3805

Event message:

- **Residual current too high**
- **Check generator**

Explanation:

Overcurrent at the DC input. The inverter briefly interrupts feed-in operation.

Corrective measures:

- If this message is displayed frequently, ensure that the PV module has been correctly rated and wired.

Event 3901, 3902

Event message:

- **Waiting for DC start conditions**
- **Start conditions not met**

Explanation:

The feed-in conditions for the utility grid are not yet fulfilled.

Corrective measures:

- Ensure that the PV module is not covered by snow or otherwise shaded.
- Wait for higher irradiation.
- If this message is displayed frequently in the morning, increase the voltage limit for starting grid feed-in. Change the parameter **Critical voltage to start feed-in**.
- If this message is displayed frequently with medium irradiation, ensure that the PV module is correctly rated.

Event 3908

Event message:

- **Leakage current-carrying capacity of DC generator too high.**

Explanation:

The permissible residual current reaches or exceeds the threshold.

Corrective measures:

- If a residual-current device is installed, ensure that it is set correctly.
- Observe the behavior of the inverter and wait to see whether the inverter resumes operation on its own.

- Ensure that no insulation error is present and that the residual-current device is functional.

Event 4011

Event message:

- **Bridged strings determined**

Explanation:

The DC inputs A and B are connected in parallel.

Event 4012

Event message:

- **No bridged strings determined**

Explanation:

The DC inputs A and B are not connected in parallel.

Event 4301

Event message:

- **Serial el.arc in String |s0| detected by AFCI mod.**

Explanation:

The inverter has detected an electric arc in the displayed string. If "String N/A" is displayed, the string could not be uniquely assigned.

Corrective measures:

- Disconnect the inverter from all voltage sources.
- Check the PV modules and the cabling in the affected string or, if the string was not displayed, in all strings for damage.

- Ensure that the DC connection in the inverter is correct.
- Repair or replace defective PV modules, DC cables or the DC connection in the inverter.
- Start manual restart (if necessary).

Event 6001 – 6499

Event message:

- **Self-diagnosis**
- **Interference device**

Explanation:

The cause must be determined by the Service.

Corrective measures:

- Contact Service.

Event 6501

Event message:

- **Self-diagnosis**
- **Overtemperature**

Explanation:

The inverter has switched off due to excessive temperature.

Corrective measures:

- Clean the cooling fins on the rear of the enclosure and the air ducts on the top using a soft brush.
- Ensure that the inverter has sufficient ventilation.
- Ensure that the inverter is not exposed to direct solar irradiation.
- Ensure the maximum ambient temperature is not exceeded.

Event 6502

Event message:

- **Self-diagnosis**
- **High pressure**

Explanation:

The inverter has switched off due to excessive pressure.

Corrective measures:

- Inspect the cooling or hydraulic system for blockages or flow restrictions.
- Verify that the pressure sensor is functioning correctly.

Event 6509**Event message:**

- **Self-diagnosis**
- **DC Bus Overvoltage**

Explanation:

The inverter has shut down because the DC bus voltage has exceeded the permissible limit.

Corrective measures:

- Check the power supply voltage and ensure it is within the rated range.
- Inspect the braking resistor or chopper circuit for proper operation.
- Restart the inverter after the voltage has stabilized. And give me an downloadable pdf file

Event 6512**Event message:**

- **Minimum operating temperature not reached**

Explanation:

The inverter will only recommence grid feed-in once the temperature has reached at least -25°C .

Event 6603

Event message:

- **Self-diagnosis**
- **Overcurrent grid (HW)**

Explanation:

The cause must be determined by the Service.

Corrective measures:

- Contact Service.

Event 6604

Event message:

- **Self-diagnosis**
- **Overvoltage intermediate circuit (SW)**

Explanation:

The cause must be determined by the Service.

Corrective measures:

- Contact Service.

Event 6801, 6802

Event message:

- **Self-diagnosis**
- **Input A defective**

Explanation:

Polarity error on the inverter.

Corrective measures:

- Check whether a string is connected to input A.
- Contact Service.

Event 6901, 6902

Event message:

- **Self-diagnosis**
- **Input B defective**

Explanation:

Polarity error on the inverter.

Corrective measures:

- Check whether a string is connected to input B.
- Contact Service.

Event 7001, 7002

Event message:

- **Fault sensor interior temperature**

Explanation:

A temperature sensor in the inverter is defective and the inverter interrupts the feed-in operation. The cause must be determined by the Service.

Corrective measures:

- Contact Service.

Event 7701, 7702, 7703

Event message:

- **Self-diagnosis**
- **Interference device**

Explanation:

The cause must be determined by the Service.

Corrective measures:

- Contact Service.

Event 8003

Event message:

- **Temperature derating**

Explanation:

The inverter has reduced its power output for more than 10 minutes due to excessive temperature.

Corrective measures:

- Clean the cooling fins on the rear of the enclosure and the air ducts on the top using a soft brush.
- Ensure that the inverter has sufficient ventilation.
- Ensure the maximum ambient temperature is not exceeded.
- Ensure that the inverter is not exposed to direct solar irradiation.

Event 8204

Event message:

- **AFCI self-test failed**

Explanation:

An error occurred during the self-test of SMA ArcFix. It is not guaranteed that SMA ArcFix will function correctly. The device does not feed in.

Corrective measures:

- Contact Service.

Event 8205**Event message:**

- **AFCI self-test successful**

Explanation:

The self-test of SMA ArcFix has been completed successfully.

Event 8708**Event message:**

- **Timeout in communication for active power limitation**

Explanation:

Communication to the system control absent. Depending on the fallback setting, either the last received values will be retained or the active power will be limited to the set percentage value of the inverter nominal power.

Corrective measures:

- Ensure that the connection to the System Manager is intact and that no cables are damaged or that no plugs have been pulled.

Event 8709

Event message:

- **Timeout in communication for reactive power spec.**

Explanation:

Communication to the system control absent.

Corrective measures:

- Ensure that the connection to the System Manager is intact and that no cables are damaged or that no plugs have been pulled.

Event 8710

Event number:

- **Timeout in communication for cos-Phi spec.**

Explanation:

Communication to the system control absent.

Corrective measures:

- Ensure that the connection to the System Manager is intact and that no cables are damaged or that no plugs have been pulled.

Event 9002

Event message:

- **Installer code invalid**

Explanation:

The SMA Grid Guard code entered is incorrect. The operating parameters are still protected and cannot be changed.

Corrective measures:

- Enter the correct SMA Grid Guard code.

Event 9003**Event message:**

- **Grid parameter locked**

Explanation:

Changes to the grid parameters are now blocked. In order to be able to make changes to the grid parameters, from now on you must log in using the SMA Grid Guard code.

Corrective measures:

- Log in as **Installer** and enter the SMA Grid Guard code.

Event 9007**Event message:**

- **Abort self-test**

Explanation:

The self-test was terminated.

Corrective measures:

- Ensure that the AC connection is correct.
- Ensure that the country data set has been entered correctly.
- Restart the self-test.

Event 10102

Display text:

Parameter |ln04| set successfully

Description:

This message is issued at the beginning of a measurement.

Event 10431

Event message:

- **I-V curve measurement was successful**

Explanation:

This message is issued at the end of a measurement.

Event 29001

Event message:

- **SMA Grid Guard code valid**

Explanation:

The entered SMA Grid Guard code is valid. The parameters protected with the SMA Grid Guard code can now be set.